

[illegible]

Meteo Station – Power Subsystem (generated)

PV→12W →> CN3791 12V MPPT →> 1S LiPo →> MT3608 (5.0V) →> DevKit 5V pin.

D1 (1N5817): MT3608 5V →> DevKit 5V pin; isolates the boost so USB VBUS wins when flashing (no back-feed).

R1/R2 divide VBAT to ~2.1V into GPIO2 (ADC1\_CH1). I2C pull-ups 4.75k on SDA/SCL.

J1

SOLAR+  
GND

PV→12W 12V 1A

U1

SOLAR+  
GND  
VBAT  
GND

CN3791 12V MPPT

U2

VBAT  
GND  
BOOST5V  
GND

MT3608 →> 5.0V

D1

V5

BOOST5V

1N5817

U3

V5  
GND  
V3V3  
VSENSE  
SDA  
SCL

ESP32–H2–DevKitM–1

C1

B+

VBAT

GND

1S LiPo 3.7V 10Ah

C2

V5

GND

100uF

C3

V5

GND

10uF

R3

V3V3

SDA

4.75k

R4

V3V3

SCL

4.75k

FUTURE: INA219 + owned 0.1ohm shunt in series with VBAT for current/power telemetry.

R1

VBAT

VSENSE

100k

R2

VSENSE

GND

100k

U4

V3V3  
GND  
SDA  
SCL

BMP388 0x76

U5

V3V3  
GND  
SDA  
SCL

MLX90614 0x5A

C5

V3V3

GND

100nF

C4

V3V3

GND

10uF

C6

V3V3

GND

100nF

Capacitor placement (physical):

C1 22uF – bulk at the MT3608 input (by the battery / boost VIN)

C2 100uF – bulk on the 5V rail, at the DevKit 5V pin

C3 10uF – beside C2 at the DevKit 5V pin

C4 10uF – bulk on the 3V3 rail, at the DevKit 3V3 pin

C5 100nF – across BMP388 VCC–GND, hard against the chip’s pins

C6 100nF – across MLX90614 VCC–GND, hard against the chip’s pins

Rule: a 100nF sits right at each chip; bulk caps go where current steps (boost out, radio, regulator).

Sheet: /  
File: meteo\_power.kicad\_sch

Title: Meteo Station Power

Size: A3Date:

Rev: 1.0Id: 1/1

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The following components are defined in the project:

- J1: PV=12W 12V 1A
- C1: BT1 1S LiPo 3.7V 10Ah
- C2: 100µF
- C3: 10µF
- C4: 10µF
- C5: 100nF
- C6: 100nF
- D1: 1N5817
- R1: 100k
- R2: 100k
- R3: 4.75k
- R4: 4.75k

FUTURE: INA219 + owned 0.1ohm shunt in series with VBAT for current/power telemetry.

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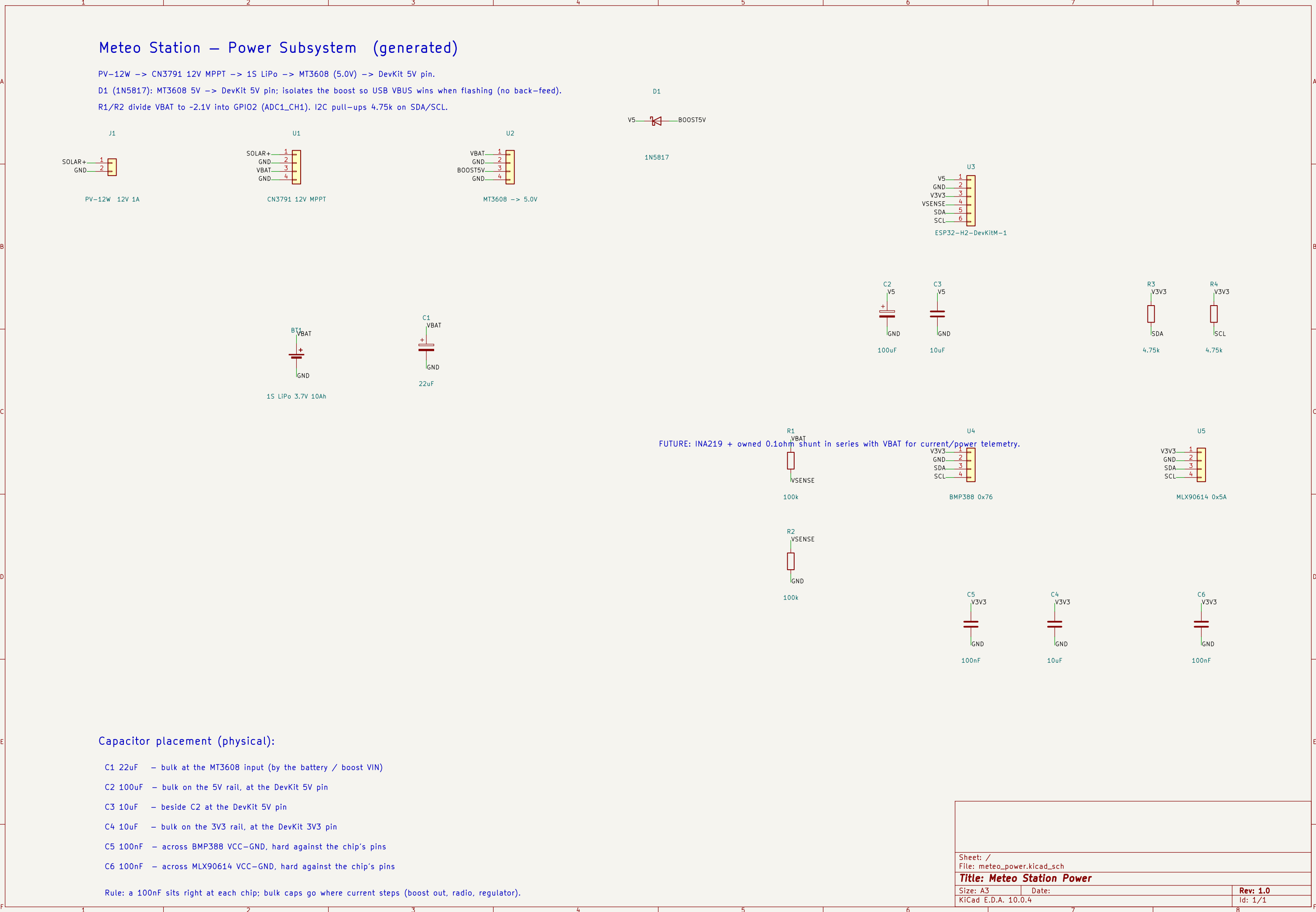
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The schematic shows the power flow from PV-12W through a CN3791 12V MPPT module to a 1S LiPo battery. The battery's output goes through a BT module and a C1 capacitor before entering the MT3608 DCDC converter. A diode D1 (1N5817) connects the MT3608 output to the V5 pin of the ESP32-H2-DevKitM-1. The V5 pin also has capacitors C2 and C3 connected to ground. The board includes two sensors: BMP388 and MLX90614, each with its own decoupling network consisting of resistors R1, R2, R3, R4 and capacitors C4, C5, C6. Pull-up resistors are shown at the I2C pins (SDA, SCL) for both sensors.

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